

Apollo TEMP_PRO-1 Sensor, Operations Guide

A field reference for deploying, configuring, and troubleshooting the TEMP_PRO-1 MQTT firmware (Ethernet and WiFi variants).

This revision matches firmware version 26.7.12.1. Earlier revisions of this guide described features that have since been removed (the drain detector, automatic power-mode flips, MQTT last-will, the retained-command scrub) and used an incorrect form for the data topics. If you have a printed copy dated May 2026 or earlier, discard it.

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What It Is

A self-contained air-quality sensor that wakes on a schedule (default every 30 minutes), samples particulate matter, temperature, and humidity, publishes one batch of readings to MQTT, and goes back to sleep. Plugged into PoE or USB it runs continuously and republishes every 10 seconds.

Built around an ESP32-S3 with deep sleep, a Sensirion SEN55 particulate sensor, an SHT3x temperature and humidity sensor, and a MAX17048 battery fuel gauge.

Battery life depends on the hardware revision and the power path more than on firmware settings. Measured results: units powered from the standard external battery pack run two to three months (the pack's own idle draw

dominates); units powered from a battery wired directly to the battery terminals run several times longer. The five month figure in earlier guides did not hold up under measurement.

Hardware Specs

Component	Spec
MCU	ESP32-S3, pinned at 160 MHz (down from 240 for power)
Particulates	Sensirion SEN55, mass concentration only (VOC/NOx disabled for short wakes)
Temp/Humidity	One SHT3x (probe-mountable)
Battery gauge	MAX17048 (always-on)
Network	W5500 Ethernet (PoE-capable) or WiFi (separate firmware file)
Connectivity	MQTT only. No Home Assistant API, no MQTT discovery
Power options	USB-C, PoE (Ethernet variant), battery
Sleep current	About 100 μ A on Rev 12 boards; about 4.9 mA on earlier revisions
Active current	Roughly 150 to 250 mA depending on workload

Power Modes (Battery vs Wired)

The single most important setting. Set by `Power Source` in the web UI, by the MQTT command topic `apollo/temp-pro-1/command/power_source`, or by a triple press of the boot button.

The device never changes this setting by itself. There is no drain detector and no automatic flip in either direction. If a unit is in the wrong mode, an operator put it there or a command did. This is deliberate: earlier firmware auto-flipped modes on sensor glitches and stale broker messages, and units disappeared into deep sleep or drained their packs.

`battery` mode (factory default)

- Wakes on a timer (`Sleep Duration`, default 30 min)
- Connects to MQTT, publishes one batch, sleeps again
- Forced asleep at `Max Wake Time` (default 45 s) no matter what, except while a firmware upload is in progress

- Below the critical battery threshold (default 5 percent) the next sleep is extended to `Critical Battery Sleep Duration` (default 60 min). The mode does not change; readings still publish every wake.

wired mode

- Stays awake continuously
- Republishes every `Publish Interval` seconds (default 10)
- Never sleeps, regardless of battery readings or boot loops

Deployment Checklist

Per unit, at install time

1. Flash the correct variant (`TEMP_PRO-1_MQTT_ETH.yaml` for PoE units, `TEMP_PRO-1_MQTT.yaml` for WiFi units).
2. Set `Power Source` for the installation type (wired for PoE/USB, battery for battery packs).
3. Set the `Column ID` (dropdown) or `Custom Column ID` (free text). This becomes part of the device's data topics. Avoid `+`, `#`, `/`, and spaces; the firmware replaces them with `_` and logs a warning.
4. Optionally set `Deployment Notes` (published as metadata, useful for "installed by, date, location" records).
5. Confirm data arrives at `Apollo/temp-pro-1-<column>/sensor/temperature`.
6. Check the boxed summary in the device log: power source, column ID, battery percent, and `Health: OK`.

Per fleet, before deploying

1. Confirm the broker address in `secrets.yaml` (or set the MQTT Broker override per device afterward).
 2. If devices sit on a different subnet than the broker, test one battery unit in place before installing the rest. Stateful firewalls that drop new cross-subnet connections break battery-mode publishing while wired units keep working; that is a network fix, not a firmware setting.
 3. Decide web UI credentials at commissioning and set per-device overrides if the site requires unique logins.
-

Boot Button Cheat Sheet

Gesture	Effect
Single press	Status flash. Blue = MQTT connected. Green = WiFi up, no MQTT (WiFi variant only). Yellow = not connected.
Triple press	Toggle power source. Solid red 3 s = now wired (stays awake). Solid green 3 s = now battery (will sleep).
Hold 10 s	Factory reset. Clears WiFi credentials, column ID, all overrides, all stored settings.
Any press while asleep	Wakes the device for one cycle.

A 20 ms debounce filter is applied in firmware, so a single scratchy press cannot register as a triple press.

Web UI Reference

Reach the web UI while the device is awake: `http://<device-ip>/` or via mDNS (`apollo-tmp-pro-1-mqtte-<mac>.local` for Ethernet units, `apollo-temp-pro-1-mqtt-<mac>.local` for WiFi units). Battery units are only reachable during a wake window; press the boot button to wake one, then triple-press into wired mode if you need time to work.

The UI is served from the device itself. It does not fetch anything from the internet, so it works on isolated networks. Sign-in uses the site's standard credentials (set at build time from `secrets.yaml`); per-device overrides are available under MQTT and Network.

Entities are grouped on the page as follows.

Live Readings

Temperature (F), Relative Humidity, PM buckets (0.3-1, 1-2.5, 2.5-4, and 4-10 μm plus total, in $\mu\text{g}/\text{m}^3$), Battery level, Battery voltage.

Calibration

Entity	Default	Notes
SHT Temperature Offset	0.0 F	Added after the C-to-F conversion. Range -10 to +10.
SHT Humidity Offset	0.0 %	Result clamped to 0-100. Range -20 to +20.
Warm-up Delay	20 s	SEN55 stabilization time before sampling. 20 s is the datasheet minimum.

Sleep and Power

Entity	Default	Notes
Power Source	battery	The master mode select. Never changes by itself.
Sleep Duration	30 min	Applies from the next sleep. Range 1-1440.
Max Wake Time	45 s	Hard cap per wake on battery. Below 45 risks losing publishes on slow networks.
Critical Battery Threshold	5 %	0 disables.
Critical Battery Sleep Duration	60 min	Used while below the threshold.
Publish Interval	10 s	Wired mode republish cadence.
Battery LED on Wake	off	Battery-level color flash on each wake. Costs battery.
Force Sleep (button)		Flips to battery mode and sleeps immediately.

MQTT and Network

Entity	Default	Notes
MQTT Broker / MQTT Port	empty / 0	Override the compile-time broker. Applies on next reboot. If an override cannot reach a broker for five consecutive boots, alternate boots retry the compile-time default so the device stays recoverable. The firmware never erases the override.
MQTT Username / Password	empty	Override compile-time credentials. Reboot to apply.
MQTT Topic Prefix	empty	Empty means <code>Apollo/temp-pro-1</code> . Changing it moves the whole data tree on the next publish, and the firmware clears the retained values at the old prefix automatically.
MQTT QoS	1	0, 1, or 2 for all data publishes. Leave at 1 unless you know why.
Web Auth Username / Password	empty	Override the built-in web login. Applies on next reboot. A forgotten password means USB reflash or the MQTT <code>web_auth</code> commands, so verify before rebooting.
Use Static IP + Static IP/Gateway/Subnet/DNS 1/DNS 2	off, DNS empty	Applied at boot on both variants. If you set a static IP and your broker is a hostname, you must set the site's DNS servers; public DNS (8.8.8.8) is usually blocked on plant networks and cannot resolve internal names. Simplest safe choice: set the MQTT Broker override to the broker's IP address.

Identification

Column ID (dropdown), Custom Column ID (free text, wins over the dropdown), Deployment Notes. All three are also settable per device over MQTT (see the topic reference).

Maintenance

ESP Reboot, Publish Now (wired diagnostics), Clean SEN55 Fan (about 10 s of fan boost, safe anytime), Buzzer Enabled (master switch), Buzzer Volume, Test Buzzer, Stop Buzzer, Restart SEN55 (recovers a wedged sensor without a reboot), Test LED, Verbose Logging (DEBUG until toggled off; resets to on at reboot).

Diagnostics (read-only)

Health Status (OK / DEGRADED / CRITICAL), Wake Reason (timer / button / cold_boot), Uptime (this wake), Device ID (MAC), IP Address, ESP Temperature, per-sensor fail counters (consecutive; reset on a good read), lifetime battery min/max, Critical Battery (this wake), RGB Light, Accessory Power (leave this on; it powers the sensors).

MQTT Topic Reference

Two different topic shapes exist, and the difference matters:

- **Data topics** (device publishes): `Apollo/temp-pro-1-<id>/...` Capital A. The device id is joined with a hyphen, so the whole thing is a single topic level: `Apollo/temp-pro-1-J8/sensor/temperature`.
- **Command topics** (device subscribes): `apollo/temp-pro-1/command/...` Lowercase a. Commands are fleet-wide broadcast except the per-device and class-scoped forms listed below.

`<id>` is the Custom Column ID if set, else the Column ID, else the last six hex digits of the MAC in lowercase (for example `352bc8`).

All data topics are retained and published at the configured QoS (default 1): battery units once per wake, wired units every publish interval.

Sensor data: `Apollo/temp-pro-1-<id>/sensor/`

Topic	Payload
<code>temperature</code>	Degrees F, one decimal. <code>-1.0</code> when the SHT3x read failed; check <code>status/sensors_skipped</code> to distinguish a failure from a real below-zero reading.
<code>humidity</code>	Percent, one decimal. <code>-1.0</code> on failure.
<code>battery_level</code>	Integer percent. <code>-1</code> on failure.
<code>pm_0_3_to_1</code> , <code>pm_1_to_2_5</code> , <code>pm_2_5_to_4</code> , <code>pm_4_to_10</code> , <code>pm_total</code>	$\mu\text{g}/\text{m}^3$, one decimal. The PM block is skipped as a whole when the SEN55 read failed; the old retained values remain, and <code>status/sensors_skipped</code> says <code>pm</code> .

Metadata: `Apollo/temp-pro-1-<id>/meta/`

`device_id` (MAC), `device_name`, `ip_address`, `firmware_version`, `boot_count`, `wake_reason`, `wake_duration_s`, `battery_voltage`, `free_heap_kb`, `deployment_notes`, `sen55_serial`, `sen55_firmware`, `power_source` (battery/wired), `sleep_mode` (legacy ON/OFF mirror of `power_source`), `sen55_fail_count`, `sht_fail_count`, `max17048_fail_count`, `lifetime_min_battery_pct`, `lifetime_max_battery_pct`, `lifetime_min_battery_v`, `lifetime_max_battery_v`, units under `unit_pm`, `unit_battery`, `unit_temperature`, `unit_humidity`, and mirrored configuration under `config_sleep_duration_min`, `config_max_wake_time_s`, `config_critical_battery_threshold_pct`, `config_critical_battery_sleep_min`, `config_warmup_delay_s`, `config_publish_interval_s`.

Status: `Apollo/temp-pro-1-<id>/status/`

Topic	Payload
health	OK , DEGRADED (a sensor failing 1-4 reads in a row or recent unexpected reboots), or CRITICAL (critical battery, a sensor failing 5+ in a row, or boot-loop cooldown). Recovered devices return to OK on their own.
sensors_skipped	Empty when the last publish was complete, else a CSV from pm , sht , batt .
critical_battery	OK or TRIGGERED .
sen55_errors	Hex status register; 0x00000000 is healthy. Not republished when the read fails, so it can go stale; cross-check the fail counters.

There is no availability or last-will topic. A device that stops publishing keeps its last retained values at the broker. To detect a dead or unreachable device, alarm on the age of meta/boot_count or meta/wake_duration_s changes in your SCADA tags (for a 30 min sleep, alarm at 90 min of silence).

Commands (all devices subscribe): apollo/temp-pro-1/command/

Topic	Payload	Notes
power_source	wired or battery (legacy ON =wired, OFF =battery)	See the retained-message guard below.
sleep_duration	minutes, 1-1440	
max_wake_time	seconds, 30-180	
warmup_delay	seconds, 0-60	
publish_interval	seconds, 1-3600	
critical_battery_threshold	percent, 0-50	
critical_battery_sleep	minutes, 5-1440	
publish_now	any non-empty payload	Wired devices only; battery devices ignore it (they publish every wake anyway).
restart	any non-empty payload	Guarded: rejected and cleared if it arrives within 60 s of the device connecting (a retained replay). Send non-retained to wired devices.
battery_led_on_wake	ON / OFF	Empty payloads ignored.
light/on , light/off , light/red , light/green , light/blue	ignored	
light/rgb	r,g,b 0-255	
light/effect	Slow Pulse , Fast Pulse , none	
buzzer	preset (beep , siren , success , error) or RTTTL string	Silent when Buzzer Enabled is off.
buzzer/stop	any	
column_id	a column from the dropdown list	Broadcast form sets EVERY device; prefer the per-device form below.
custom_column_id	free text	Same warning.
deployment_notes	free text	Same warning.

Topic	Payload	Notes
<code>mqtt_broker</code> , <code>mqtt_port</code> , <code>mqtt_username</code> , <code>mqtt_password</code>	value	Reboot to apply. Broadcast by design; the fleet shares broker credentials.
<code>web_auth_username</code> , <code>web_auth_password</code>	value	Reboot to apply. Mind the lockout risk.

Class-scoped forms exist for `power_source` and `sleep_duration`: `apollo/temp-pro-1/battery/command/...` reaches only devices currently in battery mode, `apollo/temp-pro-1/wired/command/...` only wired ones.

Per-device commands: `apollo/temp-pro-1/<mac6>/command/`

`column_id` , `custom_column_id` , and `deployment_notes` also listen on a per-device topic where `<mac6>` is the lowercase 6-hex MAC suffix (as shown in `meta/device_id` , without colons). Always use these for identity changes. Broadcasting an identity command collapses every device onto one topic prefix.

Numeric payload validation

All numeric commands reject non-numeric payloads and values outside the listed ranges, and log what they rejected. Values set by command persist across reboots and power cuts.

The retained-message guard (read this before scripting commands)

The broker replays retained command payloads to every device on every connect. Two guards protect the fleet:

- A `battery` payload on any `power_source` topic arriving within 60 seconds of that device's MQTT connect is rejected AND the retained message is deleted at the broker. Battery units are only connected during brief wakes, so anything they receive right after connecting is almost always a stale replay, and a stale `battery` used to put wired devices to sleep. To switch a connected wired device to battery: publish non-retained after it has been connected for over a minute.
- A `restart` arriving within 60 seconds of connect is likewise rejected and cleared, so a leftover retained restart cannot reboot-loop the fleet.

A retained `wired` IS accepted immediately (it is the safe direction and it is how you catch sleeping battery units; see the workflows below). Clean it up afterward: leaving it retained will flip every future battery unit to wired as it connects, and nothing will ever flip it back.

Safety Features

1. **Wake-time cap (battery mode).** A watchdog interval forces sleep at `Max Wake Time` seconds regardless of MQTT state, with a fresh sensor read and a best-effort publish first. It fires once per wake, pauses while a firmware upload is in progress (with a 10 minute upper failsafe), and cannot be defeated by anything except switching to wired mode.
2. **Critical battery.** Below the threshold, battery units extend their next sleep. Wired units only publish the flag. No mode changes.
3. **Boot-loop cooldown.** Three consecutive unexpected resets (crashes, watchdogs, brownouts; commanded reboots do not count) put a battery unit into a 30 minute cooldown sleep. Before sleeping it publishes `status/health = CRITICAL` and its wake reason so you can see why it went quiet. Wired units stay awake and show CRITICAL health instead. The counter clears after one healthy cycle.
4. **Retained-command guards.** See the topic reference above.
5. **Firmware update self-check.** After an update, the new image must connect to the broker and publish once before it is accepted. If it cannot, the device automatically rolls back to the previous firmware on the next boot. A botched update cannot brick a remote unit; it also means you should confirm one successful publish after each OTA before calling it done.
6. **Broker override fallback.** A broker/port override that has never produced a connection for five consecutive boots makes the device retry the compile-time broker on alternate boots. Recovery without a truck roll; the override itself is kept.
7. **Topic sanitization.** Illegal MQTT characters in the column ID or topic prefix are replaced with `_` at publish time (a `#` in a topic would otherwise make the broker drop the connection on every publish).

Common Operator Workflows

Deploy a new battery unit

1. Flash, power on. The unit is in battery mode with a 45 s wake cap, so do the initial config promptly, or triple-press into wired mode while you work and triple-press back when done.
2. Set Column ID, confirm data on the broker, done.

Deploy a wired (PoE/USB) unit

1. Flash, power on, triple-press (or set Power Source) to wired. Solid red confirms.
2. Set Column ID. It now publishes every 10 seconds.

Wake up all battery devices for maintenance

```
mosquitto_pub -h BROKER -t 'apollo/temp-pro-1/command/power_source' -m 'wired' -r
```

Retained, so each battery unit flips to wired as it wakes over the next sleep interval and then stays awake. When finished, clean up in this order:

```
# 1. remove the retained landmine
mosquitto_pub -h BROKER -t 'apollo/temp-pro-1/command/power_source' -n -r
# 2. wait until the devices have been connected over 60 s, then
mosquitto_pub -h BROKER -t 'apollo/temp-pro-1/command/power_source' -m 'battery'
```

(Non-retained battery command, and only after 60 s connected, or the guard rejects it. Repeat once if a unit was mid-reconnect.)

Update firmware on a battery unit

1. Flip the fleet (or the unit) to wired as above.
2. `esphome upload TEMP_PRO-1_MQTT.yaml` or upload the .bin in the web UI. The wake cap pauses during the transfer.
3. Watch for one successful publish from the new firmware. That publish is what accepts the update (safety feature 5); without it the unit reverts.
4. Flip back to battery and clear the retained command.

Find unhealthy units

```
# Health rollup for the whole fleet. Note the wildcard position: the
# device id lives INSIDE the first topic level.
mosquitto_sub -h BROKER -v -t 'Apollo/+/status/health'

# Which sensors are being skipped fleet-wide
mosquitto_sub -h BROKER -v -t 'Apollo/+/status/sensors_skipped'
```

Detect silent or dead devices

There is no last-will. Watch `Apollo/+/meta/boot_count` freshness in your SCADA tags and alarm when a device has not changed it for three sleep intervals.

Audit fleet configuration

```
mosquitto_sub -h BROKER -v -t 'Apollo/+/meta/config_sleep_duration_min'
mosquitto_sub -h BROKER -v -t 'Apollo/+/meta/firmware_version'
```

Rename a device (change its column)

Publish to the per-device topic:

```
mosquitto_pub -h BROKER -t 'apollo/temp-pro-1/352bc8/command/column_id' -m 'J8'
```

The device moves its data tree on the next publish and automatically clears the retained values at the old prefix, so no ghost device remains.

Fully reset a single unit

Hold the boot button 10 seconds, or use the Factory Reset control. This wipes WiFi credentials, column ID, all overrides, and lifetime statistics.

Troubleshooting

Device not publishing to MQTT

1. Wired or battery? A battery unit publishes for a few seconds every sleep interval; watch `Apollo/+/meta/boot_count` rather than expecting live data.
2. Check the device log (web UI or `esphome logs`). A red `REJECTED` line means the retained-command guard fired; that is the guard working.
3. `X of Y MQTT publishes DROPPED` in the log means the broker connection was down at publish time. Check broker reachability from that subnet. Battery units failing on a different subnet than the broker while wired units work is the known cross-subnet firewall issue; that fix belongs to the network team, not to device settings.
4. If a broker override is set (MQTT and Network group), confirm it is right. After five failed boots the device alternates with the compile-time broker and logs it loudly.

Battery dies faster than expected

- Check `meta/wake_duration_s`: healthy wakes run 30 to 40 s. Wakes pinned at Max Wake Time mean the device struggles to connect and every wake costs full price; fix the network path first.
- Check for a retained `wired` on the `power_source` command topics. A unit left in wired mode stays awake until the pack is flat; nothing flips it back automatically.
- Lifetime min/max battery and the fail counters reveal failing hardware that lengthens wakes.

Health: DEGRADED right after a reboot

Normal if a sensor glitched once; the counters are consecutive and clear on the next good read (a battery unit clears within one wake, a wired unit within about 5 minutes). Persistent DEGRADED with a climbing fail counter is a hardware signal.

Health: CRITICAL right after a firmware update

Commanded reboots do not count toward boot-loop detection, but a crash during the first boot of new firmware does, and rollback protection may have reverted the image. Check `meta/firmware_version` against what you flashed. If it reverted, the new image never managed a publish; investigate connectivity before flashing again.

Lifetime min battery shows a skewed value

It is a lifetime minimum and never normalizes on its own. Factory reset clears it (along with everything else), or ignore it and watch `battery_voltage` instead.

Boot button does not wake the device

The button wakes via GPIO0. If the device also does not respond on the network in wired mode, power-cycle it. Repeated `cold_boot` wake reasons afterward point at the boot-loop handling above.

MQTT broker IP changed

Fleet-wide (applies at each device's next boot):

```
mosquitto_pub -h OLD_BROKER -t 'apollo/temp-pro-1/command/mqtt_broker' -m '10.0.0.42' -r
mosquitto_pub -h OLD_BROKER -t 'apollo/temp-pro-1/command/restart' -m '1'
```

Battery units pick the retained broker value up on their next wake and use it from the boot after that. Clear the retained messages once `meta/ip_address` shows everyone on the new broker. If the old broker is already dead, wired units need the web UI; battery units need a button wake plus the web UI.

Device stuck in wired mode but should be battery

An operator or a retained command put it there. Check for retained payloads on all three `power_source` command topics, clear them, then send a non-retained `battery` while the unit has been connected over 60 s (or triple-press the button on the device).

Device in battery mode but should be wired

Same causes in reverse. Verify with `meta/power_source`. Send `wired` (retained is fine; it is accepted immediately) or triple-press.

Reference: Compile and Flash

```
# Compile only (no flash)
esphome compile TEMP_PRO-1_MQTT_ETH.yaml      # PoE/Ethernet units
esphome compile TEMP_PRO-1_MQTT_ETH.yaml      # WiFi units

# Flash via USB
esphome upload TEMP_PRO-1_MQTT_ETH.yaml --device /dev/cu.usbmodem*

# Flash over network (device must be awake; use wired mode)
esphome upload TEMP_PRO-1_MQTT_ETH.yaml --device <device-ip>

# Stream live logs
esphome logs TEMP_PRO-1_MQTT_ETH.yaml --device <device-ip>

# Erase all flash (full clean slate including stored settings)
esptool.py --chip esp32s3 --port /dev/cu.usbmodem* erase_flash
```

Build with ESPHome 2026.3.1 (the version the fleet is flashed with) unless a newer version has been verified on a bench unit first. A `secrets.yaml` is required next to the YAML files; copy `secrets.yaml.example` and fill it in.

Appendix: Topic Examples

```
# Everything from one device (the device id is inside the first level)
```

```
mosquitto_sub -h BROKER -v -t 'Apollo/temp-pro-1-J8/#'
```

```
# One value from all devices
```

```
mosquitto_sub -h BROKER -v -t 'Apollo/+/sensor/battery_level'
```

```
mosquitto_sub -h BROKER -v -t 'Apollo/+/status/health'
```

```
# Wake all battery devices into wired mode (remember the cleanup!)
```

```
mosquitto_pub -h BROKER -t 'apollo/temp-pro-1/command/power_source' -m 'wired' -r
```

```
# Restart all WIRED devices (non-retained; battery units reboot each wake anyway)
```

```
mosquitto_pub -h BROKER -t 'apollo/temp-pro-1/command/restart' -m '1'
```

```
# Set sleep duration to 1 hour for battery devices only
```

```
mosquitto_pub -h BROKER -t 'apollo/temp-pro-1/battery/command/sleep_duration' -m '60'
```

```
# Set one device's column id (per-device topic, mac suffix from meta/device_id)
```

```
mosquitto_pub -h BROKER -t 'apollo/temp-pro-1/352bc8/command/column_id' -m 'H05'
```